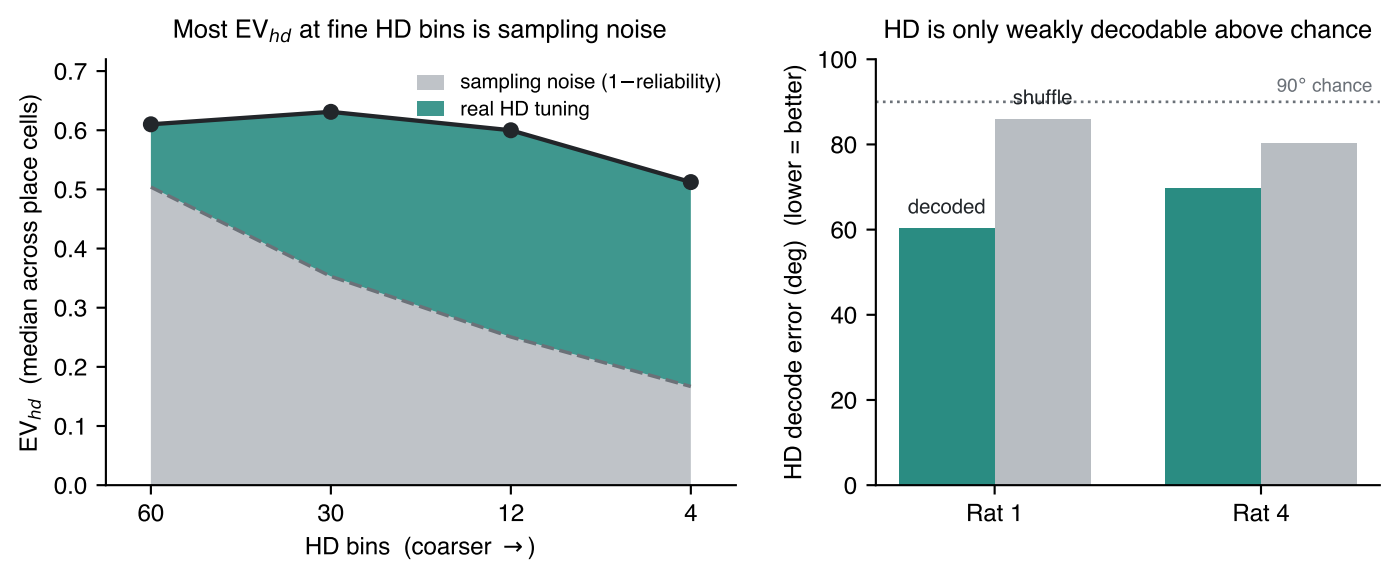
Most EV$_{hd}$ at fine HD bins is sampling noise
sampling noise (1−reliability)
real HD tuning
EV$_{hd}$ (median across place cells)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
60
30
12
4
HD bins (coarser →)
HD is only weakly decodable above chance
HD decode error (deg) (lower = better)
0
20
40
60
80
100
shuffle
90° chance
decoded
Rat 1
Rat 4